

Notes: Beshears, Choi, Clayton, Harris, Laibson, and Madrian (JFE, 2025) Optimal Illiquidity

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1 Big picture

- **Question.** What liquidity restrictions should a socially optimal retirement savings system impose when households face genuine taste shocks but also suffer from present bias?
- **Policy object.** The planner chooses mandatory balances in one or more accounts. Each account has a linear early-withdrawal penalty π_n . A liquid account has $\pi = 0$, a fully illiquid account has $\pi = 1$, and a partially illiquid account has $0 < \pi < 1$.
- **Behavioral friction.** Households are naive present-biased agents. They choose period-1 withdrawals as if the future is discounted by $\beta\delta$, while the social planner evaluates welfare without the present-bias distortion.
- **Taste-shock friction.** A private taste shock θ makes some households normatively want high period-1 consumption. The planner cannot perfectly distinguish a high- θ household with a real liquidity need from a low- β household with a self-control problem.
- **Baseline result with homogeneous present bias.** If all households have the same β , a single account with early-withdrawal penalty approximately $1 - \beta$ is almost optimal. This is the Pigouvian logic: tax early consumption by the size of the present-bias wedge.
- **Main result with heterogeneous present bias.** If β is heterogeneous and private, the social optimum is very well approximated by two accounts: one completely liquid and one completely illiquid until retirement.
- **Why two accounts?** Heterogeneity in β creates a screening problem. A single linear penalty cannot simultaneously serve high- β households that value flexibility and low- β households that benefit from commitment. A liquid-plus-illiquid pair approximates a kinked optimal budget set.
- **Three-account result.** Adding a 401(k)-like partially illiquid account generates tiny extra welfare gains. The optimized penalty is around 13%, close to the U.S. 10% early-withdrawal penalty for IRAs and 401(k)s.
- **Leakage result.** In the model's optimal three-account system, leakage from the partially illiquid account is high: around 90% in the benchmark. Under a calibration matched to U.S. account allocations, leakage falls to about 40%, close to U.S. empirical estimates.
- **U.S. policy implication.** The types of accounts in the U.S. system are approximately right: liquid balances, a 401(k)-style partially illiquid account, and fully illiquid Social Security-like wealth. The allocation across them is not: the model implies more wealth should be placed in completely illiquid accounts.
- **Economic takeaway.** A very simple system can approximate the welfare of a fully general nonlinear mechanism. The right design is not “maximum liquidity” and not “all forced savings”; it is a small menu of accounts that combines flexibility and commitment.

2 Data, calibration, and measurement

2.1 This is a quantitative mechanism-design paper, not an empirical causal paper

The paper does not estimate treatment effects from a natural experiment. It builds a two-period behavioral model and compares the welfare generated by different savings-account systems. The quantitative exercise asks how much welfare is generated by simple account systems relative to autarky and relative to more general mechanisms.

Interpretation: The paper's credibility comes from transparent primitives, mechanism-design logic, and robustness checks, not from quasi-random empirical variation. The figures and tables are numerical evaluations of optimal mechanisms under calibrated preference distributions.

2.2 Households, periods, and consumption

The economy contains a continuum of households. There are two periods:

- period 1 is working life;
- period 2 is retirement.

A household consumes $c_1(\theta, \beta)$ in period 1 and $c_2(\theta, \beta)$ in period 2. The dependence on θ and β is often suppressed in the notation.

Interpretation: The two-period abstraction allows the paper to focus on the central tradeoff between pre-retirement liquidity and retirement commitment. It deliberately abstracts from many life-cycle details.

2.3 Taste shocks

The taste shock θ shifts period-1 utility. A household with high θ has a genuine, welfare-relevant reason to consume more in working life. Examples include medical costs, housing repairs, or other urgent expenses.

Interpretation: Liquidity is valuable because some households have real period-1 needs. A fully illiquid system can hurt those households by forcing too much retirement consumption.

2.4 Present bias

Households have present-bias parameter β . A lower β means a stronger tendency to overconsume in period 1 from the perspective of the planner and the household's long-run self. The discount function is conceptually $\{1, \beta\delta, \beta\delta^2, \dots\}$.

Interpretation: Illiquidity is valuable because it limits early overconsumption by present-biased households. The planner wants to correct self-control mistakes without destroying too much flexibility.

2.5 Naivete

The model assumes households are naive about their present bias. They do not anticipate future self-control mistakes. Therefore, in autarky they do not voluntarily impose the right amount of illiquidity on themselves.

Interpretation: The assumption of naivete is crucial. If households were fully sophisticated and could commit themselves costlessly, the case for mandatory illiquidity would be weaker.

2.6 Account system

The planner offers N accounts with balances $(x_n)_{n=1}^N$ and early-withdrawal penalties $(\pi_n)_{n=1}^N$. The household can withdraw ω_n from account n in period 1. If it withdraws from account n , it receives only $(1 - \pi_n)\omega_n$ for period-1 consumption. Remaining balances are consumed in period 2.

Interpretation: An N -account system is a piecewise-linear convex budget set. Because households choose withdrawal order, they naturally withdraw first from the low-penalty account and later from higher-penalty accounts.

2.7 Welfare metric

Welfare gains are reported relative to an autarky benchmark with one completely liquid account. The paper reports gains as an equivalent percentage increase in household resources, a money-metric welfare measure.

Interpretation: A welfare gain of 6% means that the mechanism delivers welfare equivalent to raising the resource endowment by 6% under autarky.

2.8 Benchmark calibration

The benchmark simulations use log utility, normalize $R = \delta = Y = 1$, use distributions for θ and β , and evaluate account systems against the upper bound generated by the Relaxed Problem. Heterogeneous present bias is a central benchmark: β is distributed on $[0.2, 1]$.

Interpretation: The results are not an empirical estimate of the exact U.S. optimal policy. They are disciplined numerical demonstrations that simple account systems can be nearly optimal under plausible behavioral primitives.

3 Models and theoretical specifications

3.1 Household utility

Household period-1 decision utility is

$$\theta u_1(c_1) + \beta \delta u_2(c_2). \quad (1)$$

The planner evaluates the same allocation using

$$\theta u_1(c_1) + \delta u_2(c_2). \quad (2)$$

Interpretation: The difference between household choice utility and planner welfare is exactly the present-bias term β . The planner views the household's period-1 self as overvaluing immediate consumption when $\beta < 1$.

3.2 Consumer problem in an N -account system

Given account balances and penalties, the period-1 self chooses withdrawals $(\omega_n)_{n=1}^N$ to solve

$$\max_{(\omega_n)_{n=1}^N} \theta u_1(c_1) + \beta \delta u_2(c_2), \quad (3)$$

where

$$c_1 = \sum_{n=1}^N (1 - \pi_n) \omega_n, \quad (4)$$

$$c_2 = R \sum_{n=1}^N (x_n - \omega_n). \quad (5)$$

Interpretation: The penalty changes the effective exchange rate between period-2 resources and period-1 consumption. A higher π_n makes it more expensive for the present-biased self to consume retirement resources early.

3.3 Planner problem for N -account systems

The planner chooses balances and penalties to solve

$$\max_{(x_n)_{n=1}^N, (\pi_n)_{n=1}^N} \iint [\theta u_1(c_1(\theta, \beta)) + \delta u_2(c_2(\theta, \beta))] dF(\theta) dG(\beta) \quad (6)$$

subject to household optimality and the aggregate resource constraint

$$\iint \left[c_1(\theta, \beta) + \frac{c_2(\theta, \beta)}{R} \right] dF(\theta) dG(\beta) \leq Y. \quad (7)$$

Interpretation: The planner controls only the account system. It cannot directly set each type's consumption unless the account system induces those choices.

3.4 Autarky benchmark

Autarky is the reference case with one completely liquid account:

$$x_1 = Y, \quad \pi_1 = 0. \quad (8)$$

Interpretation: In autarky, the government does not impose illiquidity or transfers. Because households are naive, autarky does not generate adequate self-commitment.

3.5 General nonlinear mechanism

The General Nonlinear Mechanism allows the planner to choose arbitrary nonlinear budget sets, subject to incentive compatibility and feasibility. It is the feasible theoretical optimum and yields welfare W^G .

Interpretation: This benchmark is far richer than real-world account systems. It provides a high bar: if simple accounts come close to W^G , then policy does not need a complicated nonlinear menu to generate most welfare gains.

3.6 Relaxed Problem

The Relaxed Problem drops the monotonicity constraint from the General Mechanism while retaining the local incentive-compatibility constraint. Its welfare is W^R . Because it drops a necessary constraint, it gives an upper bound.

Interpretation: Since $W^N \leq W^G \leq W^R$, if an N -account system comes close to W^R , it must also come close to the feasible optimum W^G .

3.7 No-transfer AWA benchmark

The paper revisits Amador, Werning, and Angeletos (2006). AWA disallow inter-household transfers: penalties collected from one household cannot be rebated to others. Under homogeneous β , AWA find that a liquid account plus mandatory minimum savings is optimal.

Interpretation: The current paper generalizes AWA in two important ways: collected penalties can be rebated, and β can be heterogeneous and private.

3.8 Homogeneous present bias with transfers

When all households have the same β , the quasi-linear model yields an exact Pigouvian result:

$$\pi = 1 - \beta. \quad (9)$$

Interpretation: If the planner knows every household's present-bias wedge, a single tax on early withdrawals can directly correct it. The account system does not need to screen different β types.

3.9 Heterogeneous present bias and the inference problem

With private θ and private β , the planner observes only withdrawal choices. A high withdrawal can mean either:

- a high θ household with a real period-1 need; or
- a low β household overconsuming because of present bias.

In the quasi-linear case, the relevant revealed variable is

$$\phi = \frac{\theta}{\beta}. \quad (10)$$

The optimal marginal penalty becomes an average Pigouvian correction:

$$\pi(\phi) = E[1 - \beta \mid \phi]. \quad (11)$$

Interpretation: The planner cannot impose the exact correction $1 - \beta$ type by type. It sets a penalty based on the average present-bias wedge among types making the same revealed choice.

3.10 Pooling and kinks

When the support of β is wide relative to the support of θ , the paper obtains a pooling region. In that region, the planner cannot separate types well. The optimal budget set can feature a discrete jump in marginal penalties.

Interpretation: A liquid account plus an illiquid account is a simple way to approximate such a kinked budget set. This explains why a two-account system performs so well under heterogeneous present bias.

3.11 Three-account system and the 401(k) analogy

The paper studies a three-account system:

- account 1: completely liquid, $\pi_1 = 0$;
- account 2: partially illiquid, $0 < \pi_2 < 1$;
- account 3: completely illiquid, $\pi_3 = 1$.

The optimized middle penalty is around 13%, close to the 10% early-withdrawal penalty in U.S. IRAs and 401(k)s.

Interpretation: The model rationalizes why real-world systems may contain a partially illiquid account, but it also shows that this account contributes very little extra welfare once a liquid-plus-illiquid pair exists.

3.12 Calibrated U.S. system

The paper maps the stylized accounts to U.S. institutions:

- completely liquid wealth resembles cash and ordinary savings;
- partially illiquid wealth resembles 401(k)s and IRAs;
- completely illiquid wealth resembles Social Security or defined-benefit pensions.

The model calibrated to existing U.S. allocations generates leakage around 40.7%, in line with U.S. leakage estimates.

Interpretation: The U.S. account types are approximately right, but the model says the U.S. has too little completely illiquid saving relative to the social optimum.

4 Identification and credibility of the argument

4.1 What identifies the mechanism

The analysis is identified by theory: household preferences, private information, the government's feasible account instruments, and welfare comparisons. The key contrast is between mechanisms of increasing flexibility:

$$W^N \leq W^G \leq W^R. \quad (12)$$

Interpretation: The paper does not claim a causal estimate. It claims a welfare approximation result: simple accounts closely approximate the welfare upper bound under the specified model.

4.2 Why the upper bound matters

The Relaxed Problem is not necessarily feasible because it drops monotonicity. However, it is easier to compute and gives an upper bound. If a simple account system is close to this upper bound, the welfare lost from restricting attention to simple accounts must be small.

Interpretation: This is the main quantitative logic of the paper. The authors do not need to solve every feasible nonlinear mechanism exactly if the simple mechanism is already very close to the relaxed upper bound.

4.3 Robustness dimensions

The paper tests whether the central findings survive changes in:

- the coefficient of relative risk aversion γ ;
- the dispersion and mean of taste shocks θ ;
- the dispersion and mean of present bias β ;
- observable type heterogeneity;
- welfare weights;
- correlation between θ and β ;
- endogenous income in the appendix.

Interpretation: The robustness checks show that the two-account result is stable in the calibrated environments. The optimized middle penalty typically stays around 9–16%, and leakage from the partially illiquid account remains high.

4.4 Limitations

The paper is explicit about strong assumptions:

- only two periods are modeled;
- present bias is modeled in one specific way;
- households are naive;
- distributions of θ and β are calibrated rather than directly measured;
- account balances are imposed by government fiat in the stylized model;
- the rich tax, loan, hardship-withdrawal, rollover, and employer-match details of actual retirement systems are simplified.

Interpretation: The paper should be read as a theory-guided welfare benchmark rather than a complete blueprint for U.S. retirement policy.

5 Fully detailed section-by-section notes

5.1 Section 1: Introduction

The introduction motivates the paper using a policy puzzle. Almost all developed countries have mandatory savings that are fully illiquid before retirement, such as Social Security or compulsory pension systems. The United States is unusual because much of its defined-contribution saving is voluntary and partially liquid: 401(k)s and IRAs allow some withdrawals, often with a 10% penalty. The paper frames this institutional variation as a welfare question: is leakage from retirement savings bad, useful, or sometimes optimal?

The paper emphasizes that liquidity has two competing effects. Liquidity allows households to respond to real shocks and urgent needs. But liquidity also allows naive present-biased households to undo retirement saving. The planner must choose how much commitment to impose without eliminating valuable flexibility.

The authors preview three nested policy classes. First, N -account systems are simple, realistic mechanisms with finite accounts and linear withdrawal penalties. Second, general nonlinear mechanisms allow arbitrary budget sets and are theoretically richer. Third, the Relaxed Problem drops monotonicity and gives an upper bound on welfare. The central claim is that very simple account systems come close to the welfare of fully general mechanisms.

5.2 Section 2: Model

Section 2 lays out the two-period environment. Period 1 is working life and period 2 is retirement. Households differ in θ , a period-1 taste shifter, and β , the degree of present bias. The planner observes distributions but not individual types. Households are naive about present bias.

The account system is introduced as a set of mandatory deposits and penalties. In period 0, the planner sets account balances and penalties. In period 1, households choose withdrawals. In period 2, they consume remaining resources. Penalties collected in period 1 are rebated lump sum in equilibrium through the aggregate budget constraint.

The planner’s objective is utilitarian and excludes present bias. This creates two motives for intervention. First, the planner wants to reallocate resources intertemporally for households who overconsume in period

1. Second, the planner can use penalties and rebates to redistribute across households, although private information limits how much redistribution can be achieved.

5.3 Section 2.6: Autarky reference case

Autarky is one liquid account with no penalty. The household stores resources and chooses its own timing. Because households are naive, autarky does not produce voluntary commitment. This benchmark is used throughout the paper to report money-metric welfare gains.

5.4 Section 2.7: AWA no-transfer benchmark

The authors revisit the Amador–Werning–Angeletos model. AWA assume homogeneous present bias and no inter-household transfers. With no transfers, collected penalties cannot be rebated, so the key policy is mandatory minimum savings. This result helps situate the current paper: the current paper adds inter-household transfers and heterogeneous private present bias.

5.5 Section 3: Homogeneous present bias with transfers

Section 3 studies the case where β is homogeneous. In the quasi-linear case, the optimal penalty is exactly $\pi = 1 - \beta$. This is a Pigouvian tax: early withdrawals are taxed by the wedge between the period-1 self's preferences and the planner's welfare criterion.

With strictly concave utility, the exact formula no longer holds perfectly because redistribution matters. Still, Table 1 and Figure 1 show that the optimal penalty is very close to $1 - \beta$. A single account obtains nearly all welfare gains. Adding a second account or allowing the Relaxed Problem adds very little.

5.6 Section 4.1: Heterogeneous present bias with quasi-linear utility

Once β is heterogeneous and private, the simple Pigouvian result breaks down. A household's withdrawal choice summarizes both θ and β , making it impossible to identify the exact present-bias wedge for each household. The planner uses the revealed variable $\phi = \theta/\beta$, which combines taste needs and self-control problems.

Proposition 2 shows that the optimal marginal penalty is an average Pigouvian correction conditional on the revealed choice. Corollary 3 shows that, under a wide-support condition, the optimal allocation includes pooling and a jump in marginal penalties. This provides the intuition for why a two-account system with a kink can be nearly optimal.

5.7 Section 4.2: Heterogeneous present bias with concave utility

The paper then turns to log utility in both periods and solves the mechanisms numerically. Table 2 is the main result. A one-account system generates a 3.569% welfare gain, but the Relaxed Problem yields 6.145%. A two-account system generates 6.136%, essentially reaching the upper bound. A constrained two-account system with one liquid and one fully illiquid account generates 6.105%, also very close. Adding a third account generates only 6.137%.

The economic message is that heterogeneity in present bias makes screening important. The optimal mechanism resembles a kinked budget set, and two accounts are enough to approximate that kink.

5.8 Section 4.3: Comparative statics for the partially illiquid account

Figures 2–4 unpack why high penalties work. Figure 2 shows welfare by β type as the second account’s penalty rises. Low- β households gain dramatically from high penalties because the account protects them from overconsumption. High- β households value flexibility, so their welfare eventually declines, but the losses are small relative to the gains for low- β households.

Figure 3 shows penalty payments by type. Low- β households tend to pay the penalties because they withdraw from the penalized account. High- β households pay little and can benefit from rebates. Figure 4 aggregates welfare and shows that high penalties on the second account are desirable.

5.9 Section 4.4: Three-account system and U.S. analogy

The three-account system adds a middle account with optimized penalty π_2 , while fixing $\pi_1 = 0$ and $\pi_3 = 1$. The optimized middle penalty is about 13%, close to the 10% U.S. early-withdrawal penalty. However, the welfare gain from adding this account is tiny: 6.137% instead of 6.105% for the constrained two-account system.

Figure 5 shows that leakage from the middle account is high. About 76% of households fully draw down the partially illiquid account, 22% partially draw it down, and only 2% do not withdraw from it. This means the partially illiquid account functions more as a flexible auxiliary account than as the central commitment device.

5.10 Section 4.5: Calibrated model of the U.S. system

The authors calibrate the model to U.S. account allocations. They map Social Security and defined-benefit claims into the fully illiquid account, 401(k)/IRA balances into the partially illiquid account, and ordinary wealth into the liquid account. With this calibration, the model produces leakage of 40.7%, close to the empirical estimate that 40 cents flow out for every dollar contributed to DC/IRA accounts.

The main conclusion is that the U.S. has approximately the right account categories but too little completely illiquid saving. The optimized system puts 47.4% of resources into the completely illiquid account, compared with 32.0% under the calibrated U.S. policy.

5.11 Section 5: Robustness

Section 5 varies parameters and extensions. Across robustness checks, three findings persist: the liquid-plus-illiquid two-account system is very close to optimal; a third account has an optimized penalty near the U.S. 401(k)/IRA penalty; and leakage from the partially illiquid account remains high.

Table 3 varies risk aversion. Table 4 varies the distribution of taste shocks. Table 5 varies the distribution of present bias. The optimized middle penalty typically lies between 9% and 16%, and leakage ranges from roughly 80% to 99% in the optimal-account simulations.

5.12 Section 5.1: Observable types

The paper discusses how observable heterogeneity, such as income or ability, could change optimal policy. If high-ability households are more likely to have high θ , the planner may want to allow them more liquidity to preserve labor supply incentives and redistribution objectives. This extension reinforces the general lesson that optimal penalties depend on what type information is observable.

5.13 Section 6: Conclusions and directions for future work

The conclusion summarizes three benchmark findings. First, the optimum is well approximated by a two-account system with one liquid and one illiquid account. Second, if a third account is added, its penalty is slightly above 10%. Third, leakage from the partially illiquid account is high. These map naturally to liquid checking/savings accounts, Social Security, and 401(k)/IRA accounts.

The authors emphasize that the model is stylized. Future work should explore richer life cycles, retirement decumulation, alternative self-control models, sophistication, richer distributions, taxes, and endogenous income. The paper's policy message is therefore a disciplined benchmark, not the final word on retirement savings design.

6 Tables, figures, and original excerpts

6.1 Model setup and account system

Read:

- The model block shows household preferences, planner preferences, timing, and the N -account withdrawal problem.
- Period-1 utility is shifted by θ , while household choice is distorted by β .
- The planner chooses balances x_n and penalties π_n , but consumption choices are generated by the household's present-biased optimization.
- The aggregate budget constraint incorporates penalty revenues and resource feasibility.

Economic message: The core policy instrument is not a tax rate alone. It is a whole account system that changes the budget set faced by present-biased households.

2.1. Preferences of households

Preferences in period 1 are given by

$$\theta u_1(c_1) + \beta \delta u_2(c_2),$$

where θ is a stochastic taste shifter,¹³ $u_i : (0, \infty) \rightarrow \mathbb{R}$ is the period- i utility function, c_i is period- i consumption, β is the present-bias discount factor, and δ is the standard discount factor.¹⁴ Preferences in period 2 are given by $u_2(c_2)$.

2.2. Information structure

We assume households are naive: they do not anticipate present bias (see [Strotz, 1955](#); [O'Donoghue and Rabin, 1999a](#)). The assumption of naivete is broadly supported by the empirical literature (see reviews in [Ericson and Laibson, 2019](#); [Cohen et al., 2020](#)), although there are a range of results (e.g., see [Alcott et al., 2022](#)). The assumption of naivete is critical for the results of this paper because it eliminates the motive for self-commitment as well as the opportunity for government/planner screening in a hypothetical 'pre-period'.¹⁵

We assume that taste shifters, θ , and present bias, β , are private information of each household in the economy. The social planner knows the aggregate distribution of (θ, β) across households. We denote the distribution function of θ by $F(\cdot)$ and of β by $G(\cdot)$. In our analysis, we assume that θ and β are independent.¹⁶

2.3. Preferences of the social planner

The social planner and the household (with taste shifter θ) have nearly identical preferences over consumption in periods 1 and 2. The only difference is that the social planner does not normatively endorse present bias, implying that the planner's objective for a household is

$$\theta u_1(c_1) + \delta u_2(c_2).$$

The assumption that the social planner maximizes an objective without present bias, is a common assumption in the literature (AWA). The social planner chooses policies that maximize the utilitarian¹⁷ social objective:

$$\iint \left(\theta u_1(c_1(\theta, \beta)) + \delta u_2(c_2(\theta, \beta)) \right) dF(\theta) dG(\beta). \quad (1)$$

The social planner takes account of the (endogenous) equilibrium policy functions of the households, c_1 and c_2 . The social planner creates incentives that influence these policy functions, but cannot control them directly because the planner does not directly observe θ and β for each household. The social planner's mechanism uses total resources bounded by the aggregate endowment Y .

Eq. (1) implies that the planner has two motives in changing the allocations that emerge in an autarkic system. First, the planner would like to generate more savings, because only households, and not the planner, have present bias. Second, the planner would like to generate inter-personal reallocations from agents with low θ values to agents

with high θ values. The first motive is an inter-temporal reallocation (within a household) and the second motive is an inter-personal redistribution.

2.4. Timing

Time 0: The planner sets up N accounts, each with gross rate of return R , where N is a constraint discussed in the next section. Each of the N accounts is characterized by two variables: an initial allocation x_n and a linear withdrawal penalty π_n , which applies only to withdrawals in period 1 (i.e., an early-withdrawal penalty).¹⁸ If a consumer withdraws ω_n dollars from account n in period 1, the consumer actually receives $(1 - \pi_n)\omega_n$ dollars.¹⁹ A completely liquid account has $\pi_n = 0$, a partially liquid account has $0 < \pi_n < 1$, and a completely illiquid account has $\pi_n = 1$. For the planner, the choice variables are the allocations to the N accounts, $(x_n)_{n=1}^N$, and the respective early withdrawal penalties, $(\pi_n)_{n=1}^N$. The planner chooses the account allocations in a way that respects the economy's overall budget balance: $\sum_{n=1}^N x_n$ will equal Y plus the aggregate value of the early withdrawal penalties collected in equilibrium.

Time 1: Self 1 maximizes welfare from the perspective of time 1 (including present bias). This generates withdrawals from the accounts established at time 0. Consumption is $c_1(\theta, \beta)$.

Time 2: Self 2 spends any remaining funds in their accounts. Consumption is $c_2(\theta, \beta)$.

2.5. Summary of the N -account system

We begin with the consumer's problem, since consumer behavior is an input to the planner's problem. In period 1, the consumer with parameters θ and β maximizes

$$\max_{(\omega_n)_{n=1}^N} \theta u_1(c_1) + \beta \delta u_2(c_2), \quad (2)$$

where consumption is given by

$$c_1 = \sum_{n=1}^N (1 - \pi_n) \omega_n, \quad (3)$$

$$c_2 = R \sum_{n=1}^N (x_n - \omega_n). \quad (4)$$

Conditional on the policy vectors $(x_n)_{n=1}^N$ and $(\pi_n)_{n=1}^N$, this generates consumption levels $c_1(\theta, \beta)$ and $c_2(\theta, \beta)$, where we have suppressed the dependency on $(x_n)_{n=1}^N$ and $(\pi_n)_{n=1}^N$.

We assume a continuum of consumers (with measure one), so integrating over taste-parameters, θ and β , is the same as integrating over consumers. In period 0, the planner faces the problem

$$\max_{(x_n)_{n=1}^N, (\pi_n)_{n=1}^N} \iint \left(\theta u_1(c_1(\theta, \beta)) + \delta u_2(c_2(\theta, \beta)) \right) dF(\theta) dG(\beta) \quad (5)$$

subject to the constraints that (i) $c_1(\theta, \beta)$ and $c_2(\theta, \beta)$ are given by the consumer's problem (Eqs. (2)-(4)) and (ii) economy-wide budget balance is satisfied:

$$\iint \left(c_1(\theta, \beta) + \frac{c_2(\theta, \beta)}{R} \right) dF(\theta) dG(\beta) \leq Y. \quad (6)$$

In other words, the planner chooses the account allocation vector, $(x_n)_{n=1}^N$, and the penalty vector, $(\pi_n)_{n=1}^N$, to maximize social surplus (Eq. (5)) subject to the constraints that agents will exhibit present bias in

Original model setup: preferences, timing, household problem, and planner problem.

6.2 General mechanism and homogeneous- β Pigouvian result

Read:

- The general mechanism excerpt shows the planner's objective, budget constraint, local incentive compatibility, and monotonicity.
- The Relaxed Problem drops monotonicity and gives an upper bound on feasible welfare.
- Proposition 1 states that, under homogeneous present bias and quasi-linear structure, a one-account system with penalty $\pi = 1 - \beta$ is first-best efficient.

Economic message: If β is known and common, a simple Pigouvian penalty is enough. Complexity becomes valuable only when the planner must screen heterogeneous private β types.

¹³ See [Atkeson and Lucas \(1992\)](#) for use of such taste shifters. There are also other ways of modeling taste shifters. For example, consider $u(c - \theta)$, where θ is a taste shifter. This case is beyond the scope of the current paper.

¹⁴ This framework can be generalized WLOG by including a second independent taste shifter, ϕ , in the utility function.

selection of a utility pair, (v_1, v_2) , where $v_i = u_i(c_i)$.

In the General Non-linear Mechanism, the planner's problem can be expressed as that of choosing $v_1, v_2 : \Theta \rightarrow \mathbb{R}$ to maximize welfare

$$\int (\theta v_1(\theta) + v_2(\theta)) f(\theta) d\theta \quad (\text{Planner Objective})$$

subject to the resource constraint

$$\int \left(Y - C_1(v_1(\theta)) - \frac{1}{R} C_2(v_2(\theta)) \right) f(\theta) d\theta \geq 0 \quad (\text{Budget Constraint})$$

where $C_i = u_i^{-1}$, that is $c_i(\theta) = C_i(v_i(\theta))$, and the incentive-compatibility constraint, which now has two parts, namely a linear part,

$$0 = \theta v'_1(\theta) + \beta v'_2(\theta) \quad (\text{Local IC})$$

and a monotonic part,

$$0 \leq -v'_2(\theta). \quad (\text{Monotonicity})$$

(a) General mechanism and Relaxed Problem ingredients.

in both periods.

Proposition 1. *Suppose that all households have a strictly concave utility function in the first period, linear in the second period. Then the socially optimal 1-account system with a Pigouvian tax on consumption is also first-best efficient.*

This 1-account system is also first-best efficient.

(b) Proposition 1: homogeneous β Pigouvian penalty.

6.3 Table 1 and Figure 1: homogeneous present bias

Read:

- Table 1 shows that with homogeneous β , a single account captures almost all feasible welfare gains.
- For $\beta = 0.6$, the one-account system yields 2.794%, the two-account system yields 2.860%, and the Relaxed Problem yields 2.881%.
- For $\beta = 0.1$, welfare gains are very large because present bias is severe: roughly 70% of the endowment.
- Figure 1 shows that the optimized penalty is very close to $1 - \beta$, though slightly below the exact Pigouvian line in some cases because the planner also has redistributive motives under concavity.

Economic message: Homogeneous present bias is easy to correct: apply almost exactly the present-bias wedge as an early-withdrawal penalty.

Table 1
Welfare gains from four versions of the homogeneous- β model: an N -account System with $N = 1$; an N -account System with $N = 2$; the Relaxed Problem, which is the relaxed version of the General Non-linear Mechanism; an N -account System with $N = 2$ accounts that have exogenously set penalties, specifically $\pi_1 = 0$ and $\pi_2 = 1$, thereby implying that the first account is completely liquid and the second account is completely illiquid in period 1. In the columns we report welfare gains for 10 different values of β (namely 0.1, 0.2, ..., 1.0). The welfare gain is calculated as the percentage increase in household wealth that would produce the same average welfare in the autarkic case. Welfare is calculated using the planner's welfare criterion (i.e., without present bias in the welfare objective).

	Value of β				
	0.1	0.2	0.3	0.4	0.5
N -account System: $N = 1$	69.658	31.775	17.124	9.630	5.347
N -account System: $N = 2$	71.648	32.698	17.605	9.882	5.478
Relaxed Problem	71.674	32.748	17.659	9.928	5.511
N -account System: $N = 2, \pi_1 = 0, \pi_2 = 1$	71.633	32.648	17.482	9.671	5.196
	Value of β				
	0.6	0.7	0.8	0.9	1.0
N -account System: $N = 1$	2.794	1.283	0.446	0.067	0.012
N -account System: $N = 2$	2.860	1.314	0.458	0.070	0.012
Relaxed Problem	2.881	1.325	0.462	0.070	0.011
N -account System: $N = 2, \pi_1 = 0, \pi_2 = 1$	2.542	1.018	0.256	0.015	0.000

(a) Table 1: welfare gains under homogeneous β .

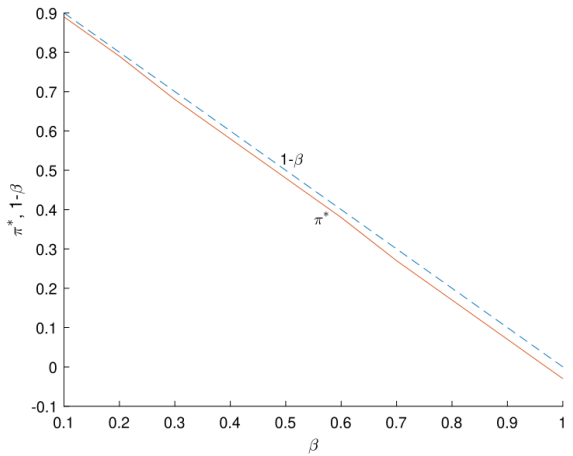


Fig. 1. The optimal penalty π^* and the notional Pigouvian tax $1 - \beta$ as a function of β in the case in which: (i) the population has homogeneous β ; (ii) the planner is confined to the N -account System with $N = 1$ accounts, with penalty π . Note that π^* is always lower than $1 - \beta$. In particular, π^* is negative at $\beta = 1$. This is due to the redistributive motive of the planner: she wishes to redistribute from types with low θ to types with high θ .

(b) Figure 1: optimized penalty versus $1 - \beta$.

6.4 Proposition 2, Corollary 3, and Table 2: heterogeneous present bias

Read:

- Proposition 2 shows that the optimal marginal penalty is an expected Pigouvian correction conditional on the revealed type ϕ .
- Corollary 3 shows that when the support of β is wide, the optimal mechanism has pooling and a jump in marginal penalties.
- Table 2 shows that one account is no longer enough: $N = 1$ generates 3.569%, while $N = 2$ generates 6.136%.
- The Relaxed Problem gives 6.145%, so the two-account system is almost at the upper bound.
- A constrained liquid-plus-illiquid system yields 6.105%, and adding a third account yields only 6.137%.

Economic message: Heterogeneous private present bias changes the nature of the optimum. The key is no longer just Pigouvian correction; it is screening, pooling, and approximating a kinked budget set.

know whether the household is making that choice because of high θ (i.e., real need), or because of low β (more present bias). The planner wants to give higher period-1 consumption to a high- θ household, but not to a low- β household.

As in the previous section, we begin with the quasi-linear case and then provide quantitative simulations.

4.1. Optimal policy with quasi-linear utility

With homogeneous β , a planner knows β even without observing ϕ . By contrast, in the heterogeneous- β case that we are now studying, the planner has to try to infer β and θ from ϕ . Because of this inference problem, exact Pigouvian taxation cannot emerge. However, Proposition 2 shows that conditionally expected Pigouvian taxation emerges as the socially optimal mechanism in a quasi-linear economy.¹⁶

Proposition 2. Suppose that inter-household transfers are possible. Assume that utility is strictly concave in the first period and linear in the second period, that the solution is interior and that $E[\theta | \phi]$ is non-decreasing in $\phi = \theta/\beta$. Then the optimal allocation is characterized by $E[\theta | \phi] u'_1(c_1(\phi)) = 1$, and the implied (local) marginal penalty rate for period 1 withdrawals is $\pi(\phi) = E[1 - \beta | \phi]$.

Corollary 3. Suppose that the assumptions of Proposition 2 are satisfied and: (i) the support of (θ, β) is β -wide and (ii) β is uniform. Then, $E[\theta | \phi]$ is constant for all $\phi \in [\theta/\beta, g/\beta]$. Consequently,

1. The optimal allocation pools all types $\phi \in [\theta/\beta, g/\beta]$.
2. There is a jump in the marginal penalty from just below the pooling region to just above the pooling region, that is,

$$\lim_{\phi \downarrow \theta/\beta} \pi(\phi) < \lim_{\phi \uparrow g/\beta} \pi(\phi). \quad (9)$$

The key driver of Corollary 3 is that the conditional expectation $E[\theta | \phi]$ is constant in ϕ over an interval, implying the existence of a pooling region. We emphasize that Corollary 3 makes an assumption about the joint support of (θ, β) and an assumption about the density of β on its support. Corollary 3 places no restriction on the density of θ on its support. The proof of Corollary 3 is in Appendix D.

Proposition 2 and Corollary 3 are expressed in terms of truth-telling (second-best) optimal mechanisms. We now address the standard problem of reinterpreting these mechanisms in terms of a system of accounts – or more generally a budget set – that represents the institutional analog of the theoretical analysis that has been provided in this subsection.

In practice, the mechanism described in Corollary 3 would be implemented institutionally as a budget set, which is a set of (c_1, c_2)

Table 2

Welfare gains from five versions of the heterogeneous- β model (with β distributed uniformly between 0.2 and 1): an N -account System with $N = 1$ (row 1); an N -account System with $N = 2$ (row 2); the Relaxed Problem (row 3), which is the relaxed version of the General Non-linear Mechanism; an N -account System with $N = 2$ accounts that have exogenously set penalties, specifically $\pi_1 = 0$ and $\pi_2 = 1$ (row 4); and an N -account System with $N = 3$ accounts, where accounts 1 and 3 have exogenously set penalties, specifically $\pi_1 = 0$ and $\pi_3 = 1$, and account 2 has an endogenous penalty (row 5).

N -account System: $N = 1$	3.569
N -account System: $N = 2$	6.136
Relaxed Problem	6.145
N -account System: $N = 2$, $\pi_1 = 0$, $\pi_2 = 1$	6.105
N -account System: $N = 3$, $\pi_1 = 0$, $\pi_3 = 1$	6.137

(a) Proposition 2 and Corollary 3: average Pigouvian correction and pooling.

(b) Table 2: welfare gains under heterogeneous β .

6.5 Figures 2–4 and Eq. (10): why the high-penalty account works

Read:

- Figure 2 shows welfare by β type as the penalty on the second account varies from 0 to 1.
- Low- β households gain dramatically from higher penalties because they need commitment.
- High- β households gain initially from transfers and later lose from reduced flexibility.
- Figure 3 shows that low- β households pay most of the penalties; high- β households pay little.
- Figure 4 shows aggregate welfare as a function of the penalty and confirms that high penalties are socially attractive.
- Eq. (10) characterizes the optimized penalty using the expected Pigouvian wedge among types who withdraw from the second account, weighted by their consumption responses.

Economic message: A high penalty creates large self-control gains for the most present-biased households. The flexibility losses for high- β households are comparatively small.

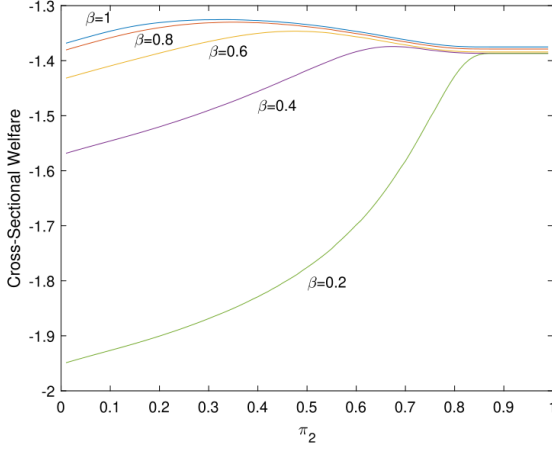


Fig. 2. The welfare of various β cross sections of the population as a function of π_2 in the case in which: (i) the population has heterogeneous β ; (ii) the planner is confined to an N -account System with $N = 2$ accounts, with penalties π_1 and π_2 respectively; (iii) $\pi_1 = 0$ (i.e., the first account is completely liquid); (iv) the account allocations are chosen to maximize the welfare of the population as a whole. Note that the sub-population with $\beta = 0.82$ (not shown) is indifferent between the system with $\pi_2 = 0$ and the system with $\pi_2 = 1$.

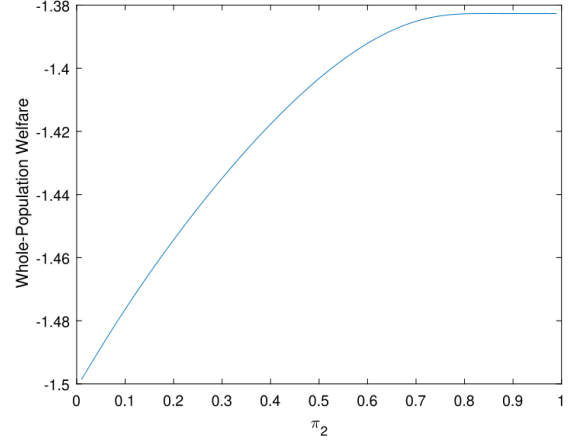
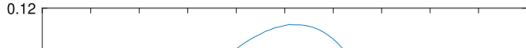


Fig. 4. The welfare of the population as a whole as a function of π_2 in the case in which: (i) the population has heterogeneous β ; (ii) the planner is confined to an N -account System with $N = 2$ accounts, with penalties π_1 and π_2 , respectively; (iii) $\pi_1 = 0$ (i.e., the first account is completely liquid); (iv) the account allocations are chosen to maximize the welfare of the population as a whole. Note that: (i) while this is not immediately apparent from the figure, the function in question is non-monotone; (ii) the optimal penalty π_2^* is approximately 85%; (iii) π_2^* yields a proportional increase of approximately 0.00002% in money-metric welfare relative to the case in which $\pi_2 = 1$ (i.e., the case in which the second account is completely illiquid). In particular, the welfare cost of setting the penalty on the second account too low far exceeds that of setting it too high.



Figures 2 and 4: welfare by β type and aggregate welfare as π_2 varies.

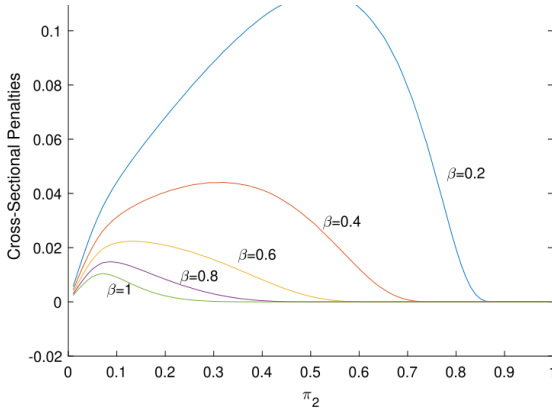


Fig. 3. The total penalties paid by various β cross sections of the population as a function of π_2 in the case in which: (i) the population has heterogeneous β ; (ii) the planner is confined to an N -account System with $N = 2$ accounts, with penalties π_1 and π_2 , respectively; (iii) $\pi_1 = 0$ (i.e., the first account is completely liquid); (iv) the account allocations are chosen to maximize the welfare of the population as a whole.

Unlike the welfare of low- β households, which rises monotonically as π_2 rises, the welfare of high- β households eventually peaks and thereafter falls with π_2 . This single-peaked property arises because, while initial rises in π_2 simply result in greater revenue from the early-withdrawal penalties paid by low- β households, later rises tend

experience a welfare loss equivalent to 0.23% of their income as the planner moves from $\pi_2 = 0$ to $\pi_2 = 1$ in Fig. 2. However, this welfare loss is swamped by the welfare gain experienced by the $\beta = 0.2$ types (which is two orders of magnitude larger).⁴⁰

Fig. 4 shows the welfare of the population as a whole as a function of the early-withdrawal penalty π_2 . It confirms that – as one would expect – the enormous welfare gains for low- β households swamp the modest welfare losses for high- β households, an example of asymmetric paternalism (Camerer et al., 2003). Although it appears that total social welfare rises monotonically and asymptotes, social welfare actually reaches a global maximum at $\pi_2 = 0.85$ and then falls very slightly. However, the fall in welfare between $\pi_2 = 0.85$ and $\pi_2 = 1$ is insignificant: it is 0.00002% of wealth using a money metric. Accordingly, the social optimum is effectively obtained with one completely liquid account and one completely illiquid account.

Optimal π_2 in the quasilinear case. To further develop intuition, we map the analysis of this section to the quasilinear case of Section 4.1. That is, we study a 2-account system in which one account is completely liquid, the second account has a penalty π_2 , and the planner optimizes over the penalty π_2 and the account allocations. In Appendix D.3, we show that the planner's optimal penalty in the quasilinear case satisfies

$$\mathbb{E} \left[\left[\mathbb{E}[1 - \beta|\phi] - \pi_2 \right] \cdot \frac{\partial c_1(\phi)}{\partial \pi_2} \mid \phi \geq \phi_2 \right] = 0, \quad (10)$$

where ϕ_2 is the lowest ϕ type that withdraws from the second account in equilibrium. Eq. (10) reveals two reasons why the optimal penalty on the second account tends to be large. The first is that the penalty

Figure 3 and Eq. (10): penalties paid by type and the first-order condition for π_2 .

6.6 Figure 5: leakage from the partially illiquid account

Read:

- Figure 5 plots the cumulative distribution of c_2/c_1 in the three-account system.
- The optimized middle account has penalty $\pi_2 = 0.13$.
- The paper reports that about 76% of households fully draw down the partially illiquid account.
- Another 22% partially withdraw from it.
- Only about 2% leave the partially illiquid account untouched.

Economic message: Leakage is not automatically a policy failure. In the benchmark system, the central commitment device is the fully illiquid account; the partially illiquid account is meant to preserve flexibility and generate modest additional screening.

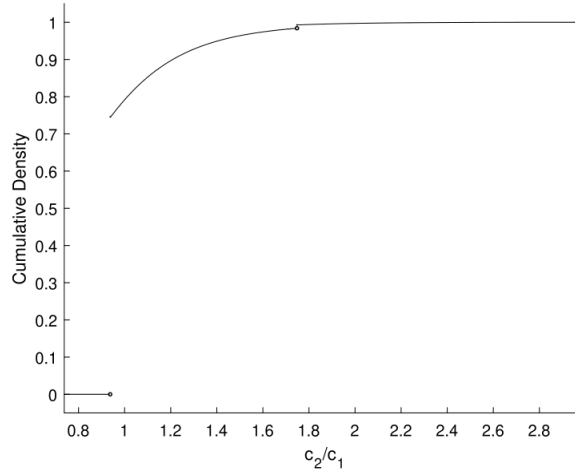


Fig. 5. The distribution function of the ratio c_2/c_1 of period-2 consumption to period-1 consumption in the population as a whole in the case in which: (i) the population has heterogeneous β ; (ii) the planner is confined to an N -account System with $N = 3$ accounts, with penalties π_1 , π_2 and π_3 , respectively; (iii) $\pi_1 = 0$ (i.e., the first account is completely liquid); (iv) $\pi_3 = 1$ (i.e., the third account is completely illiquid); (v) both π_2 and the account allocations are chosen to maximize the welfare of the population as a whole. There are two atoms in the distribution: a large atom accounting for about 76% of the total mass near $c_2/c_1 = 0.94$; and a small atom accounting for about 1% of the total mass near $c_2/c_1 = 1.70$. Individuals at the second atom have withdrawn the entire balance from the first (liquid) account, but have not yet touched the second account. Individuals at the first atom have withdrawn the entire balance from both the first and the second accounts. In particular, they have paid the penalty π_2 on the entire balance of the second account.

Figure 5: distribution of retirement-to-working-life consumption ratio in the three-account system.

6.7 Tables 3–5: robustness checks

Read:

- Table 3 varies risk aversion γ . Welfare gains shrink as risk aversion rises, but the two-account and three-account systems remain very close to the Relaxed Problem.
- Table 4 varies the dispersion and mean of the taste shock θ . The optimized middle-account penalty generally ranges from 9% to 16%, and leakage remains high.
- Table 5 varies dispersion and mean of β . When β heterogeneity disappears, the logic reverts toward the homogeneous case and a two-account fully illiquid structure is less central.

- Across calibrations, the incremental welfare from adding a third account is small.

Economic message: The main results are robust: simple liquid-plus-illiquid accounts approximate the upper bound; if a third account is added, its penalty is 401(k)-like; leakage remains a normal feature of the optimum.

Table 3
Robustness checks for welfare gains, optimal penalties and leakage rates. In each sub-table: row 1 contains welfare gains from an N -account System with $N = 2$ accounts that have exogenously set penalties, specifically $x_1 = 0$ and $x_2 = 1$; row 2 contains welfare gains from an N -account System with $N = 3$ accounts, where accounts 1 and 3 have exogenously set penalties, specifically $x_1 = 0$ and $x_3 = 1$, and account 2 has an endogenous penalty; rows 3 and 4 contain the optimal penalty and leakage rate from the endogenous-penalty account associated with the system in row 2; and row 5 contains welfare gains from the Relaxed Problem. This table varies the value of the coefficient of relative risk aversion γ .

	Value of γ			
	0.5	1.0	2.0	4.0
N -account System: $N = 2, x_1 = 0, x_2 = 1$	8.853	6.106	3.261	1.623
N -account System: $N = 3, x_1 = 0, x_3 = 1$	8.920	6.137	3.274	1.633
— — — — Penalty x_2^*	0.13	0.13	0.11	0.13
— — — — Leakage Rate	0.89	0.90	0.99	0.88
Relaxed Problem	8.935	6.145	3.279	1.634

Variation of the coefficient of relative risk aversion γ

(a) Table 3: risk-aversion robustness.

Table 5
Robustness checks for welfare gains, optimal penalties and leakage rates. In each sub-table: row 1 contains welfare gains from an N -account System with $N = 2$ accounts that have exogenously set penalties, specifically $x_1 = 0$ and $x_2 = 1$; row 2 contains welfare gains from an N -account System with $N = 3$ accounts, where accounts 1 and 3 have exogenously set penalties, specifically $x_1 = 0$ and $x_3 = 1$, and account 2 has an endogenous penalty; rows 3 and 4 contain the optimal penalty and leakage rate from the endogenous-penalty account associated with the system in row 2; and row 5 contains welfare gains from the Relaxed Problem. This table varies parameters of the distribution of β for a truncated normal distribution of β on support $[0, 2.10]$. Table 5 panel (a) varies the parameter σ_β of the truncated-normal distribution of β . panel (b) varies the parameter μ_β of the truncated-normal distribution of β .

	Value of σ_β			
	$+\infty$	1.0	0.5	0.0
N -account System: $N = 2, x_1 = 0, x_2 = 1$	6.106	6.020	5.773	2.542
N -account System: $N = 3, x_1 = 0, x_3 = 1$	6.137	6.053	5.811	2.841
— — — — Penalty x_2^*	0.13	0.13	0.14	0.36
— — — — Leakage Rate	0.90	0.90	0.90	0.73
Relaxed Problem	6.145	6.061	5.820	2.881

(a) Variation of the standard deviation σ_β of the present bias distribution

	Value of μ_β					
	0.45	0.50	0.55	0.65	0.70	0.75
N -account System: $N = 2, x_1 = 0, x_2 = 1$	6.690	6.377	6.072	5.481	5.197	4.921
N -account System: $N = 3, x_1 = 0, x_3 = 1$	6.730	6.417	6.111	5.519	5.234	4.958
— — — — Penalty x_2^*	0.15	0.15	0.15	0.14	0.14	0.14
— — — — Leakage Rate	0.91	0.90	0.89	0.88	0.87	0.85
Relaxed Problem	6.741	6.427	6.120	5.527	5.242	4.966

(b) Variation of the mean μ_β of the present bias distribution

Table 4
Robustness checks for welfare gains, optimal penalties and leakage rates. In each sub-table: row 1 contains welfare gains from an N -account System with $N = 2$ accounts that have exogenously set penalties, specifically $x_1 = 0$ and $x_2 = 1$; row 2 contains welfare gains from an N -account System with $N = 3$ accounts, where accounts 1 and 3 have exogenously set penalties, specifically $x_1 = 0$ and $x_3 = 1$, and account 2 has an endogenous penalty; rows 3 and 4 contain the optimal penalty and leakage rate from the endogenous-penalty account associated with the system in row 2; and row 5 contains welfare gains from the Relaxed Problem. This table varies parameters of the distribution of θ . Table 4 panel (a) varies the parameter σ_θ of the truncated-normal distribution of θ . panel (b) varies the parameter μ_θ of the truncated-normal distribution of θ , without changing the truncated support. panel (c) varies the parameter μ_θ of the truncated-normal distribution of θ , while moving the truncated support to maintain a symmetric truncated distribution.

	Value of σ_θ					
	0.40	0.35	0.30	0.25	0.20	0.15
N -account System: $N = 2, x_1 = 0, x_2 = 1$	5.661	5.770	5.919	6.106	6.323	6.531
N -account System: $N = 3, x_1 = 0, x_3 = 1$	5.715	5.819	5.959	6.137	6.344	6.541
— — — — Penalty x_2^*	0.16	0.15	0.14	0.13	0.12	0.09
— — — — Leakage Rate	0.80	0.84	0.84	0.90	0.89	0.99
Relaxed Problem	5.725	5.829	5.968	6.145	6.349	6.544

(a) Variation of the standard deviation σ_θ of the taste shock

	Value of μ_θ (Fixed Support)		
	0.8	1.0	1.2
N -account System: $N = 2, x_1 = 0, x_2 = 1$	6.048	6.106	6.103
N -account System: $N = 3, x_1 = 0, x_3 = 1$	6.095	6.137	6.125
— — — — Penalty x_2^*	0.15	0.13	0.12
— — — — Leakage Rate	0.83	0.90	0.88
Relaxed Problem	6.095	6.145	6.136

(b) Variation of the mean μ_θ of the taste shock holding fixed the support.

	Value of μ_θ (Moving Support)		
	0.8	1.0	1.2
N -account System: $N = 2, x_1 = 0, x_2 = 1$	5.970	6.106	6.052
N -account System: $N = 3, x_1 = 0, x_3 = 1$	6.016	6.137	6.072
— — — — Penalty x_2^*	0.15	0.13	0.11
— — — — Leakage Rate	0.82	0.90	0.95
Relaxed Problem	6.027	6.145	6.078

(c) Variation of the mean μ_θ of the taste shock, with moving support.

(b) Table 4: taste-shock robustness.

6. Conclusions and directions for future work

We focus our summary on our benchmark case in which agents have heterogeneous present bias and the planner can implement mechanisms with inter-household transfers. Three findings emerge from our analysis:

1. The constrained-efficient social optimum is well-approximated by a two-account system, with one account that is completely liquid and a second account that is completely illiquid. Little welfare gain is obtained by moving beyond this simple two-account system. Accordingly, the two-account system identified in AWA (in a model with homogeneous β and no inter-household transfers) turns out to be approximately optimal in our new setting (with heterogeneous β and inter-household transfers).
2. If a third account is added, its optimized early-withdrawal penalty is only slightly above 10%.
3. In equilibrium, the leakage rate from this (partially illiquid) third account is high. We report a range of equilibrium leakage rates, depending on the calibration. With optimal allocations to all three accounts – completely liquid, partially illiquid, and completely illiquid – equilibrium leakage rates from the partially

6.8 Conclusion excerpt

Read:

- The paper’s conclusion states three benchmark findings: two accounts approximate the constrained optimum; a third account has a penalty slightly above 10%; and leakage from that third account is high.
- These features match broad elements of the U.S. system: liquid accounts, Social Security-like illiquid wealth, and 401(k)/IRA-like partially illiquid accounts.

Economic message: The policy message is subtle: the account categories resemble real institutions, but the optimal allocation likely puts more wealth into fully illiquid saving.

7 Short summary

- The paper studies optimal retirement-account illiquidity with taste shocks and naive present bias.
- Liquidity is valuable because households have real period-1 needs; illiquidity is valuable because present-biased households overconsume early.
- The planner chooses mandatory balances and early-withdrawal penalties across accounts.
- In homogeneous- β environments, a one-account Pigouvian penalty $\pi \approx 1 - \beta$ is almost optimal.
- In heterogeneous- β environments, a one-account system performs poorly because private β creates a screening problem.
- A two-account system with one completely liquid account and one completely illiquid account nearly reaches the welfare upper bound.
- Adding a third partially illiquid account gives almost no additional welfare, but its optimized penalty is around 13%, close to the U.S. 401(k)/IRA 10% penalty.
- Leakage from the partially illiquid account is high in the benchmark optimum because that account is not the main commitment device.
- When calibrated to U.S. account allocations, the model generates about 40% leakage, matching empirical estimates.
- The U.S. system has approximately the right types of accounts, but the model implies more wealth should be accumulated in completely illiquid accounts.

8 Conclusion

8.1 Core conclusion

Beshears et al. show that optimal illiquidity can be implemented surprisingly well with a simple account system. With homogeneous present bias, a single account with a Pigouvian withdrawal penalty works well. With heterogeneous private present bias, the optimal mechanism is well approximated by two accounts: one liquid and one fully illiquid.

8.2 Economic intuition

The planner faces a commitment-flexibility tradeoff. Present-biased households need commitment, but households with high taste shocks need liquidity. When β is private, the planner cannot perfectly distinguish these motives. A liquid account protects flexibility, while a fully illiquid account protects retirement consumption. The combination approximates the kinked budget set that the optimal nonlinear mechanism would choose.

8.3 Takeaway

The paper provides a theoretical rationale for a retirement system with simple account categories: liquid wealth, fully illiquid mandatory saving, and a small partially illiquid account. It also warns that observed leakage from partially illiquid accounts is not necessarily a welfare problem. The bigger policy issue may be whether enough wealth is held in fully illiquid accounts.